

5.

The screenshot shows the Embarcadero Dev-C++ IDE with a project named "Project1". The main.c file contains the following code:

```
1 #include <stdio.h>
2
3 // Function prototype
4 int power(int base, int exp);
5
6 int main()
7 {
8     int base, exp;
9
10    // Read input
11    printf("Enter base and exponent: ");
12    scanf("%d %d", &base, &exp);
13
14    // Display result
15    printf("%d^%d = %d\n", base, exp, power(base, exp));
16
17    return 0;
18 }
19
20 // Recursive function to calculate power
21 int power(int base, int exp)
22 {
23     // Base case
24     if (exp == 0)
25     {
26         return 1;
27     }
28
29     // Recursive case
30     return base * power(base, exp - 1);
31 }
32
```

The Compiler window shows the following output:

```
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\M S I\Desktop\C\Project1.exe
- Output Size: 322.8017578125 KiB
- Compilation Time: 0.64s
```

The status bar at the bottom indicates "Line: 14 Col: 22 Sel: 0 Lines: 32 Length: 529 Insert Done parsing in 0.079 seconds".

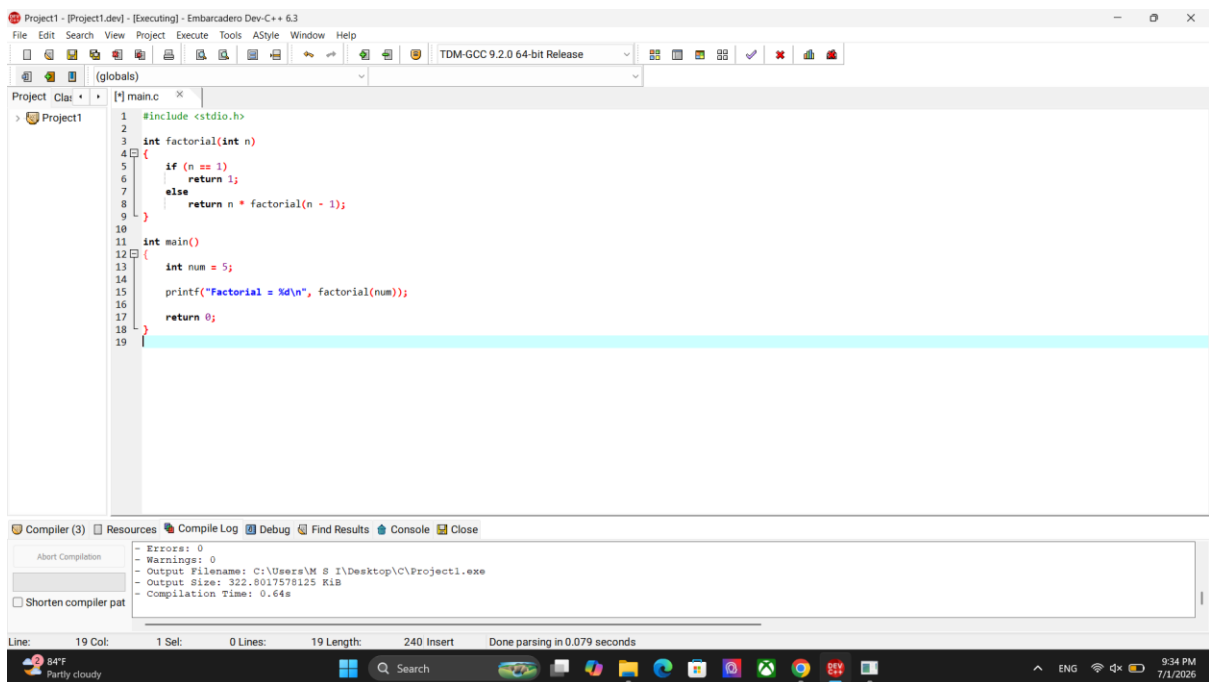
The screenshot shows the console window with the following output:

```
C:\Users\M S I\Desktop\C\Project1.exe
Enter base and exponent: 2 10
2^10 = 1024

-----
Process exited after 168.8 seconds with return value 0
Press any key to continue . . .
```

The status bar at the bottom indicates "Line: 14 Col: 22 Sel: 0 Lines: 32 Length: 529 Insert Done parsing in 0.079 seconds".

7.



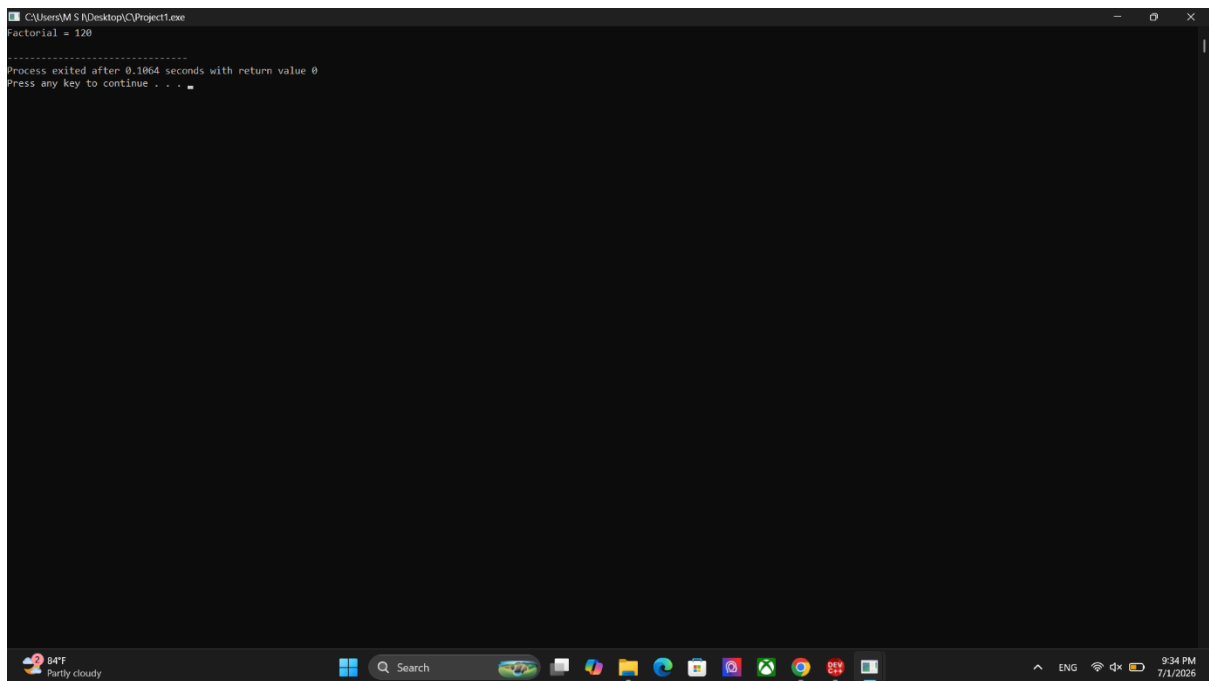
The screenshot shows the Embarcadero Dev-C++ IDE. The main window displays a C program named `main.c` with the following code:

```
1 #include <stdio.h>
2
3 int factorial(int n)
4 {
5     if (n == 1)
6         return 1;
7     else
8         return n * factorial(n - 1);
9 }
10
11 int main()
12 {
13     int num = 5;
14     printf("Factorial = %d\n", factorial(num));
15     return 0;
16 }
17
18
19
```

The bottom panel shows the Compiler window with the following output:

```
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\M S I\Desktop\C\Project1.exe
- Output Size: 322.6017578125 KiB
- Compilation Time: 0.64s
```

The status bar at the bottom indicates the current line is 19, column 19, and the file is 19 lines long. The system tray shows the date and time as 9:34 PM on 7/1/2026.



The screenshot shows a command prompt window titled `C:\Users\M S I\Desktop\C\Project1.exe`. The output of the program is displayed as follows:

```
Factorial = 120
-----
Process exited after 0.1864 seconds with return value 0
Press any key to continue . . .
```

The status bar at the bottom indicates the current line is 19, column 19, and the file is 19 lines long. The system tray shows the date and time as 9:34 PM on 7/1/2026.

9.

The screenshot shows the Embarcadero Dev-C++ 6.3 IDE. The main window displays a C++ program in a file named `main.c`. The program defines two recursive functions: `sumDigits` and `countDigits`, and a `main` function that prompts the user for a positive integer and prints the results of these functions. The code is as follows:

```
1 #include <stdio.h>
2
3 int sumDigits(int n);
4 int countDigits(int n);
5
6 int main()
7 {
8     int num;
9
10    printf("Enter a positive integer: ");
11    scanf("%d", &num);
12
13    printf("Sum of digits = %d\n", sumDigits(num));
14    printf("Number of digits = %d\n", countDigits(num));
15
16    return 0;
17 }
18
19 int sumDigits(int n)
20 {
21     if (n == 0)
22         return 0;
23     else
24         return (n % 10) + sumDigits(n / 10);
25 }
26
27
28 int countDigits(int n)
29 {
30     if (n == 0)
31         return 0;
32     else
33         return 1 + countDigits(n / 10);
34 }
35
36
```

The bottom status bar shows the compiler (TDM-GCC 9.2.0 64-bit Release) and the console output.

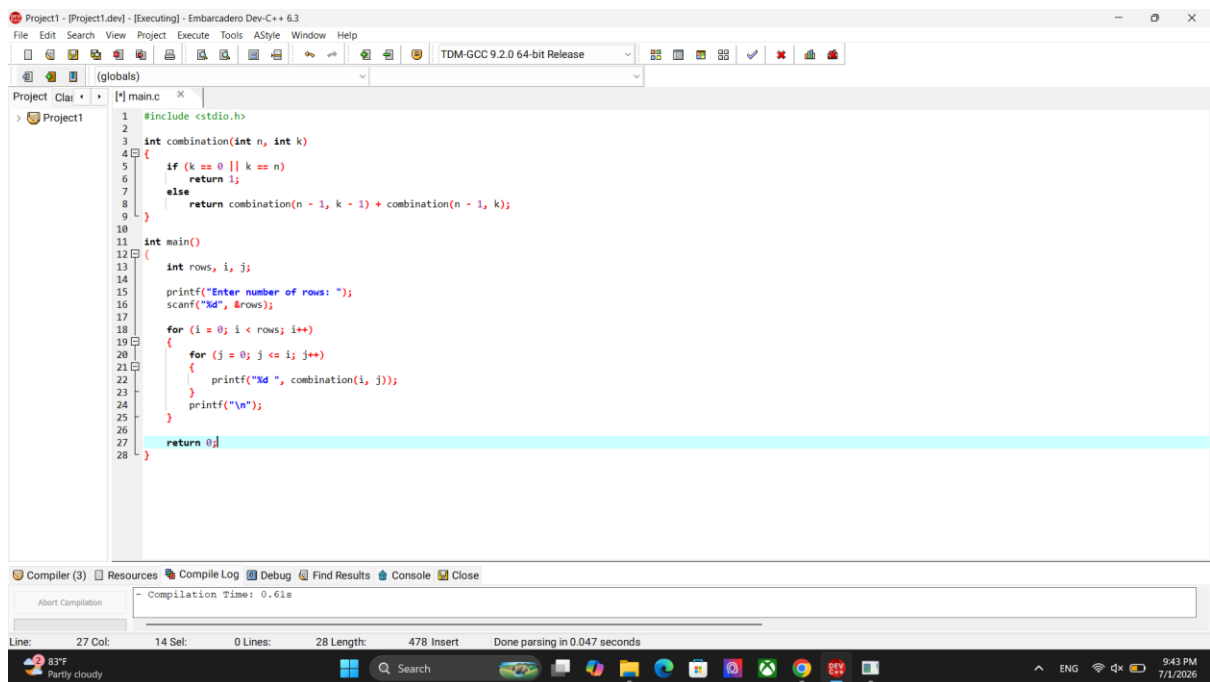
The screenshot shows the console output of the program. The user has entered the number 12345. The program has calculated the sum of digits (15) and the number of digits (5). The console output is as follows:

```
C:\Users\MS\Desktop\Project1.exe
Enter a positive integer: 12345
Sum of digits = 15
Number of digits = 5

-----
Process exited after 26.35 seconds with return value 0
Press any key to continue . . .
```

The bottom status bar shows the system clock and temperature.

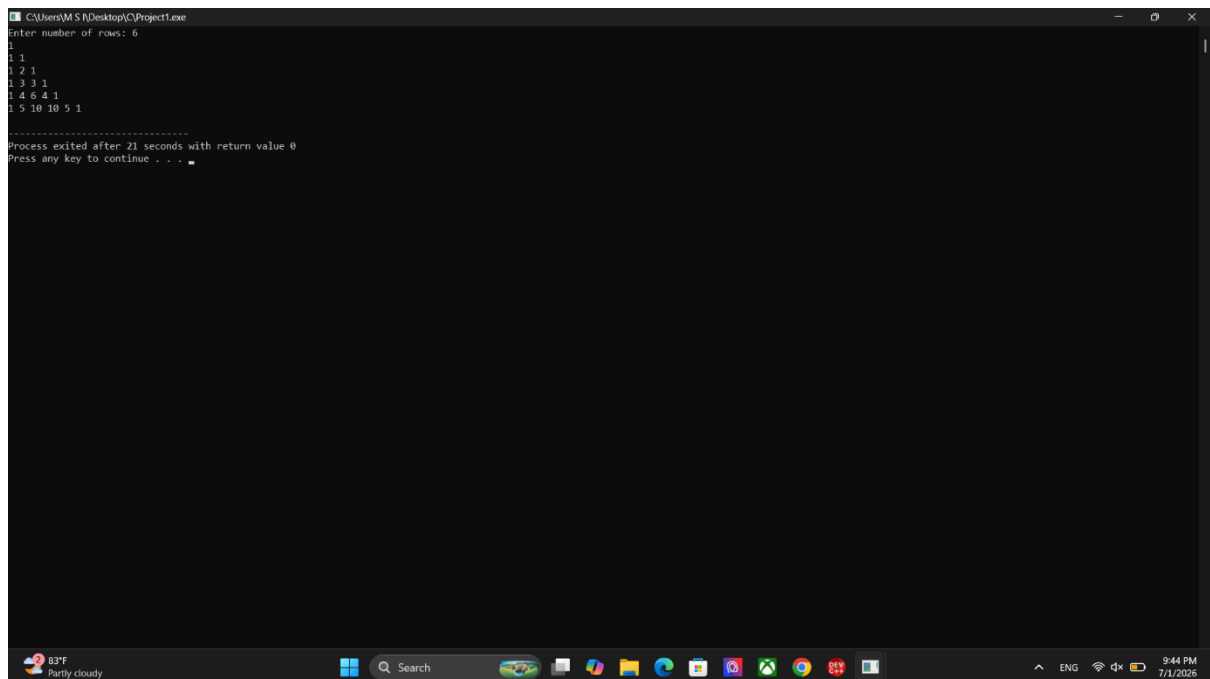
11.



The screenshot shows the EMBARCADERO Dev-C++ IDE with a project named "Project1". The code in the editor is as follows:

```
1 #include <stdio.h>
2
3 int combination(int n, int k)
4 {
5     if (k == 0 || k == n)
6         return 1;
7     else
8         return combination(n - 1, k - 1) + combination(n - 1, k);
9 }
10
11 int main()
12 {
13     int rows, i, j;
14     printf("Enter number of rows: ");
15     scanf("%d", &rows);
16
17     for (i = 0; i < rows; i++)
18     {
19         for (j = 0; j <= i; j++)
20         {
21             printf("%d ", combination(i, j));
22         }
23         printf("\n");
24     }
25
26     return 0;
27 }
```

The compiler output at the bottom shows a successful compilation with a time of 0.61s.



The screenshot shows the command prompt output of the program. The user entered 6 for the number of rows, and the program printed the following Pascal's triangle:

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
```

After the output, the program exited with a return value of 0, and the prompt asked the user to press any key to continue.